Employee Attrition Analysis

IBM HR Analytics Employee Attrition & Performance

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Introduction

Employee attrition is an indication of several underlying issues, such as poor management and communication, which are often neglected and ignoring them poses a significant and potentially fatal risk. For this project, I have explored the IBM HR Analytics Employee Attrition & Performance dataset to determine the reasons leading to employee attrition and how it can be predicted and prevented early. The analysis was performed using a spreadsheet for data cleaning, modelling and visualization. The aim is to identify the key factors that can be used by HR professionals for developing effective retention policies.

Data Interpretation & Analysis

The dataset provides a detailed and structured overview of 1,470 employees, featuring various demographic, job role and performance metrics. This data encompasses a wide range of features, including quantitative measures (e.g., salary and age) and qualitative factors (e.g., performance rating and relationship satisfaction).

Data Preprocessing

Data Cleaning: The dataset must be cleaned before moving to the next phase to avoid erroneous assumptions or misleading results.

Duplicate values: The data was checked for duplicates using conditional formatting and verified with the remove duplicates option in Google Sheets.

Missing values: Missing values were checked using conditional formatting to highlight them and use filter option in the data menu to filter and remove them.

Mapping tables: Created mapping tables using XLOOKUP, and bucketing the tenure ranges using ARRAYFORMULA.

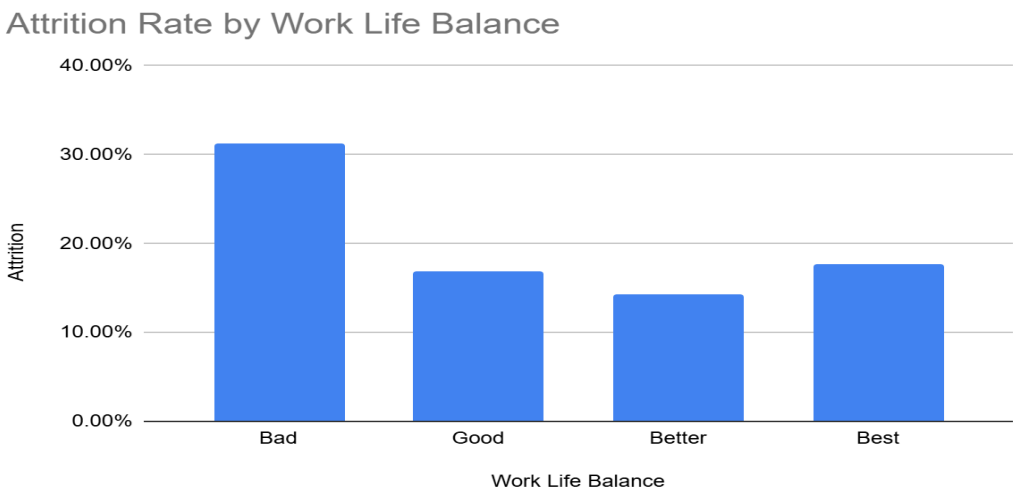
Data Exploration: Exploratory Data Analysis (EDA)

In order to find the employee attrition rates, I focused on the following key metrics:

- **Work-life Balance:** Employees reporting lower work-life balance are more likely to leave.
- **Job Satisfaction:** Employees who are content with their roles exhibit lower attrition rates.
- **Monthly Income:** Highly paid employees show less tendency to leave their roles.
- **Performance Rating:** Good performance correlates with a significantly reduced rate of employee departure.
- **Relationship Satisfaction:** Stronger relationships with the workplace correlate with a reduced likelihood of employee departure.
- **Years at Company:** A reduced number of years at the company correlates with increased employee attrition.

Data Analysis & Visualizations

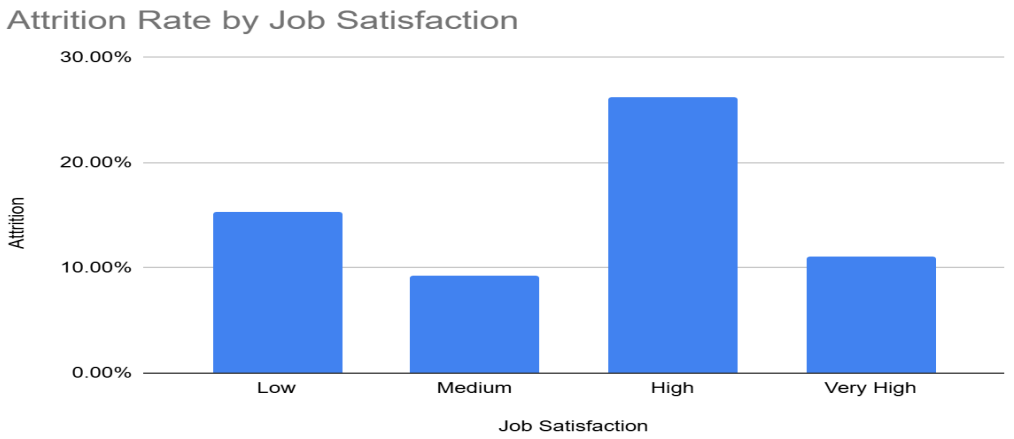
1. Work-Life Balance vs Attrition



Comparison of Employee Attrition Percentages Across Work-Life Balance Tiers.

Work-life balance is a critical factor in attrition, employees with the best work-life balance have a higher attrition rate of 17%, which is nearly half the 31% rate of those with poor work-life balance, demonstrating its crucial role in employee retention.

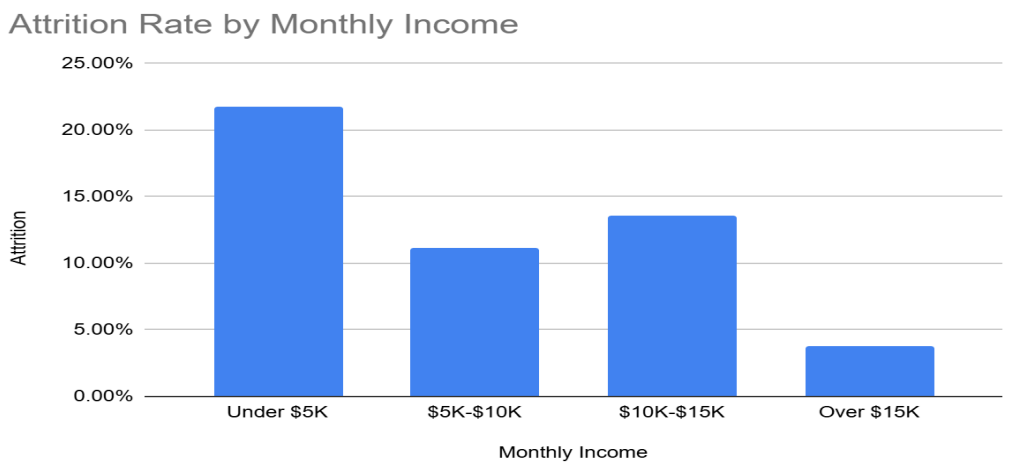
2. Job Satisfaction vs Attrition



The impact of Job Satisfaction on Overall Retention.

Employees who report a high level of job satisfaction (26%%) show a higher attrition rate than those with low satisfaction (15%) or very high satisfaction (11%). This trend suggests that for this group, turnover is driven by a lack of personnel growth, skill development and engagement, rather than a source of distress.

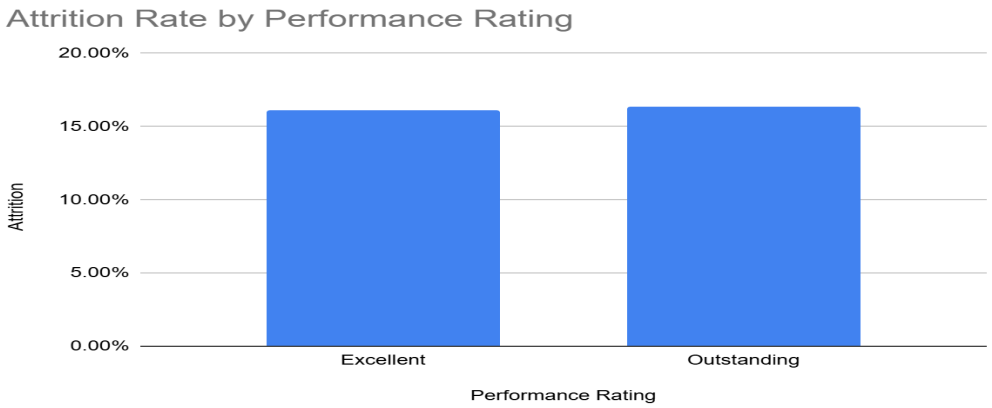
3. Monthly Income vs Attrition



Employee Attrition Rates Across Monthly Income Ranges.

Analysis of monthly income and attrition reveals a strong inverse correlation: as income levels increase, the likelihood of employee departure significantly decreases. Employees in lower income brackets exhibit a substantially higher attrition rate compared to their higher-earning counterparts, suggesting that salary and financial compensation are significant drivers of employee retention. This factor is critical for HR policy development aimed at mitigating turnover among the lowest-paid segments of the workforce.

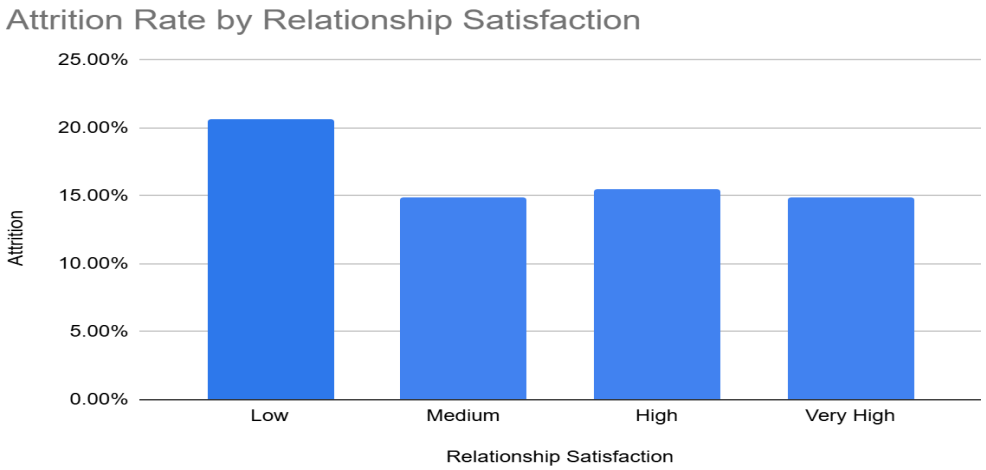
4. Performance Rating vs Attrition



The effect of performance on Employee Attrition

Employee performance shows a clear inverse relationship with attrition: high-performing employees (rated 'Excellent' or 'Outstanding') have a much lower attrition rate than those with 'Good' or 'Low' performance ratings. This suggests that recognition, career development, and appropriate challenging work correlated with strong performance are effective retention tools. Conversely, higher turnover among lower-rated employees may indicate issues with fit, support, or motivation, which should be addressed through targeted training and performance management strategies.

5. Relationship Satisfaction vs Attrition

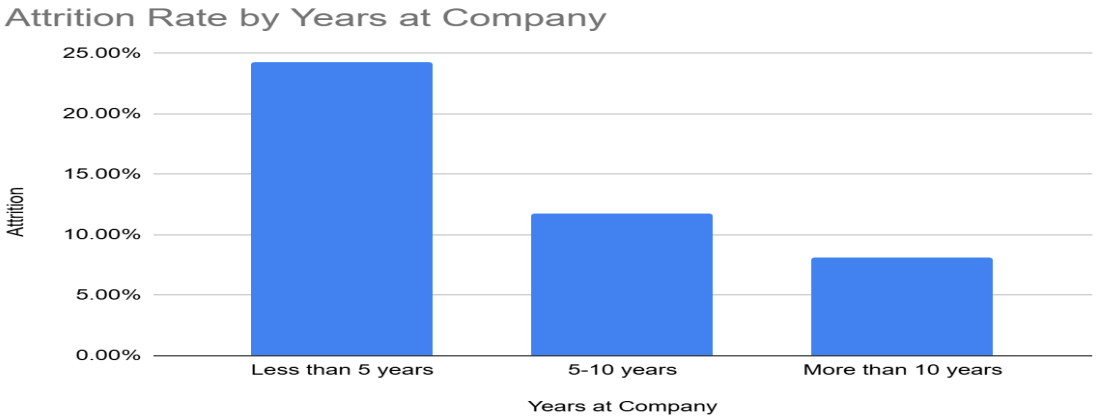


The Influence of Relationship Satisfaction on Employee Attrition

Relationship satisfaction plays a significant role in an employee's decision to stay or leave the company. Employees reporting the highest levels of relationship satisfaction exhibit the lowest attrition rates (approximately 11%). Conversely, those with the lowest relationship satisfaction levels show a

substantially higher attrition rate (around 20%). This indicates that positive interpersonal relationships, particularly with peers and managers, are a powerful retention factor, and fostering a supportive and harmonious work environment is crucial for mitigating employee turnover.

6. Years at Company vs Attrition



Employee Attrition Rates Based on Years at Company

Analysis of employee attrition by tenure clearly indicates that employees in the early stages of their career at the company (typically 1–3 years) have the highest attrition rates. This trend suggests that new hires or those who have not yet fully integrated into the company culture or career path are the most vulnerable to turnover. As tenure increases, the attrition rate generally declines, stabilizing at a much lower percentage for long-serving employees. This highlights the importance of robust onboarding, early career development, and consistent engagement strategies during an employee's first few years to maximize retention.

Key Insights

Based on the analysis of the IBM HR Analytics Employee Attrition & Performance dataset, the following key insights were derived:

- **Inverse Correlation with Financial & Performance Factors:** High Monthly Income and strong Performance Ratings are the most effective retention tools, showing a clear inverse relationship with attrition. Employees in lower income brackets and those with lower performance ratings are significantly more likely to leave.
- **Early Career Vulnerability:** Tenure is a major predictor of attrition, with the highest turnover occurring among employees in their first three years at the company. This suggests a need for enhanced focus on onboarding and early career development/engagement.
- **Complex Relationship with Satisfaction:** While generally assumed to reduce turnover, the analysis showed a paradox with **Job Satisfaction**. Employees reporting high satisfaction (but not "Very High") exhibited a higher-than-average attrition rate, suggesting that turnover for this group is driven more by a lack of growth opportunities, skill development, or engagement rather than distress or dissatisfaction with the core role.
- **Work Environment Matters:** Both **Relationship Satisfaction** and **Work-Life Balance** are critical retention factors. Employees with poor relationship satisfaction and poor work-life balance are significantly more likely to depart. Fostering a supportive work environment and ensuring a sustainable work-life balance are essential for reducing turnover.

For HR Teams:

- Focus on robust support systems for new hires
- Prioritize competitive compensation (especially for lower-paid segments)
- Emphasize on structured performance recognition, career development paths and work-life balance initiatives.

CONCLUSION

Based on the analysis of this HR dataset, it is clear that employee attrition is not a random occurrence, but a predictable outcome tied to specific workplace variables. The data reveals that retention is most heavily impacted by work-life balance, relationship satisfaction and entry-level compensation. By directly addressing these core drivers of attrition, leadership can significantly reduce continuous turnover costs and stabilize the workforce. Ultimately, fostering a supportive, fairly compensated environment is the most sustainable path to long-term organizational success.